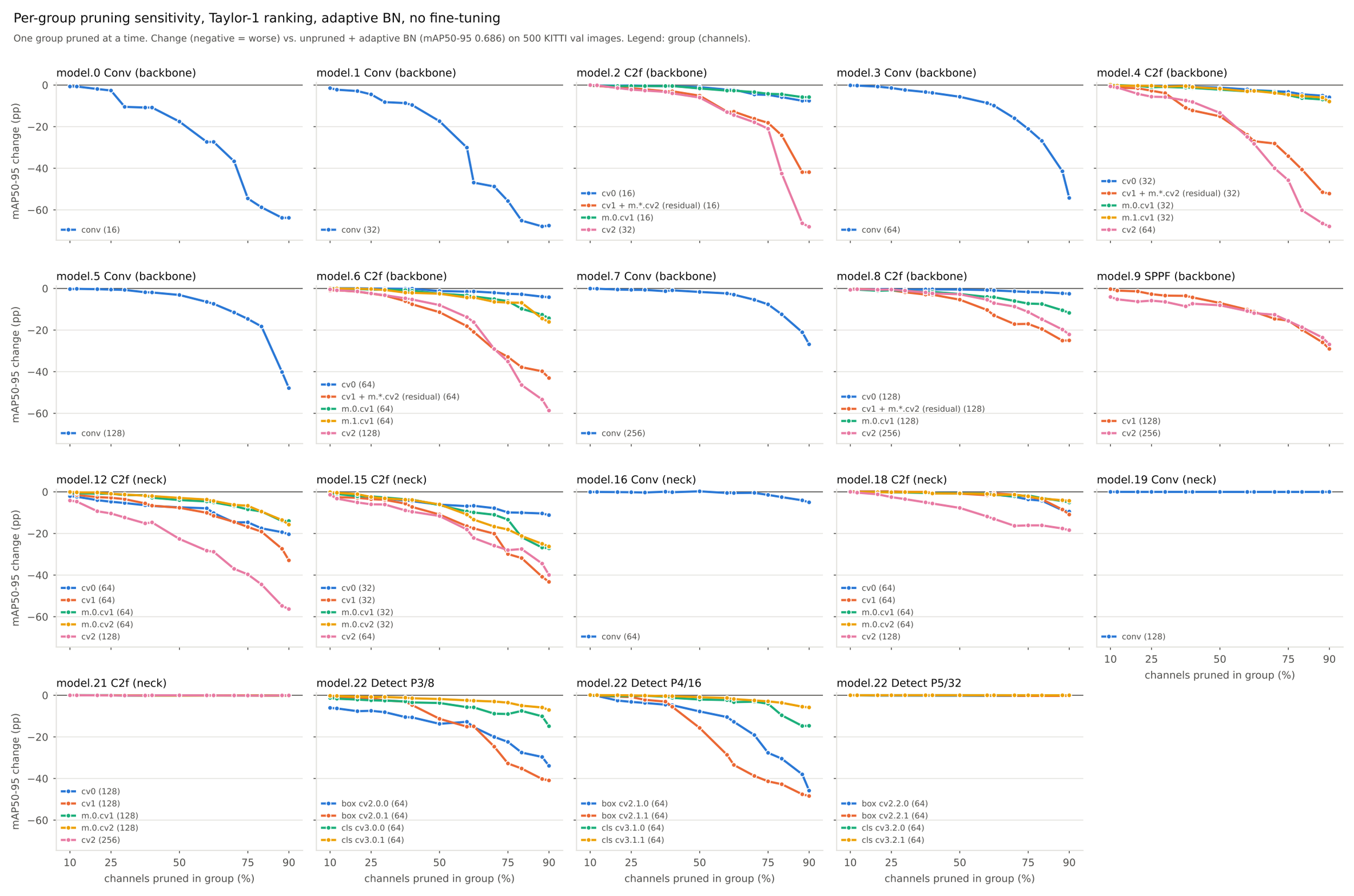

Per-group pruning sensitivity, Taylor-1 ranking, adaptive BN, no fine-tuning
One group pruned at a time. Change (negative = worse) vs. unpruned + adaptive BN (mAP50-95 0.686) on 500 KITTI val images. Legend: group (channels).
model.0 Conv (backbone)
conv (16)
model.1 Conv (backbone)
conv (32)
model.2 C2f (backbone)
cv0 (16)
cv1 + m.*.cv2 (residual) (16)
m.0.cv1 (16)
cv2 (32)
model.3 Conv (backbone)
conv (64)
model.4 C2f (backbone)
cv0 (32)
cv1 + m.*.cv2 (residual) (32)
m.0.cv1 (32)
m.1.cv1 (32)
cv2 (64)
model.5 Conv (backbone)
conv (128)
model.6 C2f (backbone)
cv0 (64)
cv1 + m.*.cv2 (residual) (64)
m.0.cv1 (64)
m.1.cv1 (64)
cv2 (128)
model.7 Conv (backbone)
conv (256)
model.8 C2f (backbone)
cv0 (128)
cv1 + m.*.cv2 (residual) (128)
m.0.cv1 (128)
cv2 (256)
model.9 SPPF (backbone)
cv1 (128)
cv2 (256)
model.12 C2f (neck)
cv0 (64)
cv1 (64)
m.0.cv1 (64)
m.0.cv2 (64)
cv2 (128)
model.15 C2f (neck)
cv0 (32)
cv1 (32)
m.0.cv1 (32)
m.0.cv2 (32)
cv2 (64)
model.16 Conv (neck)
conv (64)
model.18 C2f (neck)
cv0 (64)
cv1 (64)
m.0.cv1 (64)
m.0.cv2 (64)
cv2 (128)
model.19 Conv (neck)
conv (128)
model.21 C2f (neck)
cv0 (128)
cv1 (128)
m.0.cv1 (128)
m.0.cv2 (128)
cv2 (256)
model.22 Detect P3/8
box cv2.0.0 (64)
box cv2.0.1 (64)
cls cv3.0.0 (64)
cls cv3.0.1 (64)
model.22 Detect P4/16
box cv2.1.0 (64)
box cv2.1.1 (64)
cls cv3.1.0 (64)
cls cv3.1.1 (64)
model.22 Detect P5/32
box cv2.2.0 (64)
box cv2.2.1 (64)
cls cv3.2.0 (64)
cls cv3.2.1 (64)
mAP50-95 change (pp)
0
−20
−40
−60
10
25
50
75
90
channels pruned in group (%)